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Instructional support on first aid in choking by an artificial intelligence-powered chatbot

Application of emerging digital technologies, including mobile devices and artificial intelligence (AI), has the potential to improve outcomes in life-threatening emergencies through engagement of bystanders into provision of first aid, and is currently in the forefront of emergency medicine research [1-3].

Recently, Microsoft Corporation (Redmond, Washington, USA) launched the new Bing – an AI-powered web search assistant with chatbot functionality. The new Bing runs on a next-generation GPT-4 large language model [4], that is declared to be even more powerful than the ground-breaking ChatGPT [5,6]. When compared to regular search engines which respond to a user's query by providing a list of relevant links, the new Bing conducts one or more web searches, and uses the top results found across the web to generate a single, summarised, and "grounded" response [5]. In March 2023, the new Bing had passed the 100 million daily active users mark [7]. As the new technology becomes extremely popular and available to almost every user, it is important to understand its implications for public health. With respect to first aid, the AI-enhanced chatbot could represent a source of immediate informational support for untrained people by giving an advice on how to help a victim in a real-life emergency situation. This study aimed to evaluate ability of the chatbot to provide life-saving instructions in response to a user's query concerning first aid in choking.

On April 4th, 2023, the new Bing chatbot was repeatedly queried in English with the question "What to do if someone is choking?" using the Microsoft Edge desktop web browser. Search region was set as the United Kingdom. A total of 20 queries were submitted. Chatbot responses and web articles (indicated by the chatbot as a source) were assessed for conformance with current guidelines on first aid in choking [8-10] using a checklist. Additionally, the chatbot responses were evaluated for readability based on the Flesch-Kincaid Grade Level [11].

Results of the assessment are shown in Table 1 and available in online repository [12]. The analysis found that the new Bing chatbot outperforms previously tested question-answering systems (QAS) [13-15] in terms of the ability to address user's request concerning first aid. Firstly, in the analysed sample of responses, the new Bing chatbot always provided a query-specific answer containing some instructions on first aid in choking, whereas this was not the case for voice-based conversational assistants which commonly did not recognise users' questions [13,14]. Secondly, when compared to the Google web search engine's QAS [15], the chatbot demonstrated significantly better coverage of the guidelines-compliant instructions on first aid, and much more accurate suggestion of the region-specific emergency medical services telephone number (90.0% responses instructed to call the United Kingdom's national emergency number 999). Further, the chatbot never instructed to perform actions contradictory with current first aid guidelines, that could create risks for causing harm to a victim. Based on the readability analysis, the chatbot responses are fairly easy to read and understand by the general audience: median readability amounted 5.4, meaning that a user requires a US 5–6 grade level of education (age 10–12 years old) to understand the text.

However, the chatbot responses were variable in quality and oftimes omitted potentially life-saving instructions on first aid (see Table 1). In particular, 30.0% and 45.0% responses did not explain how to perform back blows and abdominal thrusts, respectively; only one response (5.0%) instructed to continue alternating between five back blows and five abdominal thrusts if choking is not relieved after five initial abdominal thrusts; and no response recommended to commence cardiopulmonary resuscitation if a choking person becomes unresponsive. Along with that, when a response contained pertinent first aid instructions, the instructions commonly did not satisfy the checklist criteria completely (see Table 1), and therefore were considered partially compliant with the guidelines. The quality issues could be explained on one hand by inability of the AI algorithm to recognise the full spectrum of highest-priority provisions on the first aid within the source articles to include them into a summarised response and on the other hand by weaknesses of the source articles themselves. Although being published by acknowledged entities and having significantly higher compliance with the first aid guidelines compared with the chatbot responses (see Table 1), most source articles demonstrated incomplete adherence to the guidelines, that is in line with previous studies indicating that information on first aid found online is commonly having quality flaws [16-19].

Considering the aforementioned limitations, for now the new Bing chatbot cannot be recognised as a reliable source of information on first aid in choking. It should be acknowledged that the chatbot developer recommends users to utilise their own judgment and double check the facts before taking action based on Bing's responses [20]. However, taking into account the steadily growing trend for searching health information online [21] and the breakthrough rise in the popularity of newly developed AI-powered chatbots with easy-to-use interfaces like ChatGPT and the new Bing, it seems likely that the general public will increasingly use the cutting-edge QAS to support their decisions and consider them as most trusted sources of information despite developers' warnings and disclaimers. Integration of the chatbots into mobile devices, that is already the case for the new Bing, increases the probability that an individual will turn online to use the AI assistant to find an answer quickly and easily.

When it comes to the sensitive topic of first aid the informational support should be evidence-based, concise, accurate, and clearly understandable. In our opinion, in order to make the advice effective and avoid potential harm from misinformation, developer of the studied chatbot and developers of alternative AI-powered QAS [13-15] should ban provision of the variable and generally low in quality AI-generated responses to the first aid queries, and seek for cooperation with authoritative research organisations and experts to develop and implement a uniform guidelines-concordant comprehensive knowledge base on first aid, which would serve as a source of reliable and

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Table 1
Distribution of the chatbot responses and source articles in terms of satisfaction of the guidelines-based checklist criteria.

Checklist criteria	Completely satisfied		Partially satisfied		Completely or partially satisfied	
	Chatbot responses ($N = 20$), % (n)	Source articles ($N = 6$), % (n)	Chatbot responses ($N = 20$), % (n)	Source articles ($N = 6$), % (n)	Chatbot responses ($N = 20$), % (n)	Source articles ($N = 6$), % (n)
1. Does the response/source article instruct to encourage the victim to cough in mild choking, when the victim is able to cough?	50.0% (10)	66.7% (4)	50.0% (10)	16.7% (1)	100.0% (20)	83.3% (5)
2. Does the response/source article instruct to give up to five back blows when the victim shows signs of severe choking (cannot cough, speak or breathe) and is conscious?	0.0% (0)	33.3% (2)	100.0% (20)	66.7% (4)	100.0% (20)	100.0% (6)
3. Does the response/source article describe correct technique for back blows (to lean the victim forwards and apply firm blows between the shoulder blades with the heel of one hand)?	45.0% (9)	83.3% (5)	25.0% (5)	16.7% (1)	70.0% (14)	100.0% (6)
4. Does the response/source article instruct to give up to five abdominal thrusts (Heimlich maneuver) if the back blows are unsuccessful?	60.0% (12)	83.3% (5)	40.0% (8)	16.7% (1)	100.0% (20)	100.0% (6)
5. Does the response/source article describe correct technique for abdominal thrusts (to stand behind the victim, lean the victim forwards, put the fist between their ribs and their belly button and pull it sharply inwards and upwards using the other hand)?	5.0% (1)	66.7% (4)	50.0% (10)	33.3% (2)	55.0% (11)	100.0% (6)
6. Does the response/source article instruct to immediately access EMS if the abdominal thrusts are unsuccessful?	90.0% (18)	66.7% (4)	5.0% (1)	33.3% (2)	95.0% (19)	100.0% (6)
7. Does the response/source article instruct to continue alternating between five back blows and five abdominal thrusts if choking is not relieved after five abdominal thrusts?	0.0% (0)	66.7% (4)	5.0% (1)	33.3% (2)	5.0% (1)	100.0% (6)
8. Does the response/source article instruct to give CPR if the person becomes unresponsive?	0.0% (0)	50.0% (3)	0.0% (0)	33.3% (2)	0.0% (0)	83.3% (5)
Mean percentage	31.3	64.6*	34.4	31.3	65.6	95.8

Notes. * Significant difference ($p < 0.05$; Wilcoxon signed-rank test) between the chatbot responses and source articles. Abbreviations: CPR, cardiopulmonary resuscitation; EMS, emergency medical services.

standardised responses for just-in-time support of laypeople in emergency situations.

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CRediT authorship contribution statement

Alexei A. Birkun: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Adhish Gautam:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis.

Declaration of Competing Interest

The authors have no conflicts of interest to declare.

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